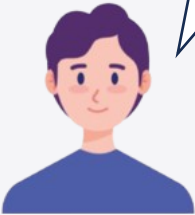


Given the phenotype “cytokine-driven hyperinflammation after viral infection”, propose 10 druggable human targets with brief rationale.



User

Orchestrate Agent (Planning & Dispatch)

Planning



1. Search literature or databases for targets linked to phenotype & druggability.
2. Extract candidate human proteins.
3. Select top 10 and provide rationales.

Dispatch

Task:

- ❑ Search scientific literature and target databases to identify human. Return a ranked list of at least 15 candidates with supporting evidence.



Retrieve Agent

Retrieve Agent (Initial Search & Analysis))

Tool Call

Query: cytokine-driven hyperinflammation viral infection human targets druggable



PubMed



UniProt

Tool Response

Analysis



Key pathways identified. (IL-6, Complement, TNF, IL-1). Broad searches need refinement for specific targets.

Retrieve Agent (Refined Search & Execution))

Tool Call

Query: "Interleukin-6 signaling", "TNF superfamily", "Complement C3/C5", "NLRP3 inflammasome"



RCSB PDB



UniProt

Tool Response

Result



Compiled List: 15+ high-confidence, druggable targets (e.g., IL6, CS, JAK1, NLRP3, TNF, CSAR1, etc.) with functional notes.

Orchestrate Agent (Review & Output)

Review



15 candidates reviewed. Top 10 selected based on strong mechanistic link and druggability.

Output



Final List of 10 Druggable Human Targets:

1. IL6 (Interleukin-6): Rationale: A master regulator cytokine whose levels correlate with disease severity in viral hyperinflammation (e.g., severe COVID-19). Direct inhibition is clinically validated.
 2. IL6R (Interleukin-6 Receptor): Rationale: The specific receptor for IL-6; blocking it disrupts classic and trans-signaling. Monoclonal antibodies like tocilizumab are approved for cytokine release syndromes.
 3. C5 (Complement C5): Rationale: A pivotal point in the complement cascade; its cleavage generates C5a (potent inflammatory mediator) and initiates the membrane attack complex (MAC). Inhibition (e.g., eculizumab) can halt complement-driven immunopathology.
- C5AR1 (C5a Receptor 1)
-